# Loss Function Quiz – Questions and Answers

**Q: What is the primary purpose of a loss function in training ML models?**

A: A loss function measures how far the model’s predictions deviate from the true target values.

**Q: How does a loss function quantitatively measure the performance of an ML model?**

A: It provides a numerical value that quantifies model prediction errors.

**Q: What are the two main types of target variables mentioned in the context of loss functions?**

A: The two main target variable types are continuous (regression) and categorical (classification).

**Q: What must be done to the loss function to obtain the best/optimal set of model parameters?**

A: To get optimal model parameters, the loss function must be minimized.

**Q: What are the two alternative names for the Mean Squared Error (MSE)?**

A: Mean Squared Error (MSE) is also called L2 Loss or Quadratic Loss.

**Q: What is the formula for $L_{MSE}$?**

A: $L_{MSE} = (1/n)\ \Sigma\ (y_i - \blacksquare_i)^2$.

**Q: Why is MSE a very widely used loss function?**

A: MSE is widely used because it is smooth, differentiable, and emphasizes larger errors.

**Q: How does MSE's penalty mechanism affect larger errors?**

A: MSE penalizes larger errors quadratically, increasing their influence.

**Q: Due to its penalty mechanism, what is MSE sensitive to in a dataset?**

A: MSE is sensitive to outliers because squared terms amplify extreme deviations.

**Q: What property of the MSE function guarantees a global minimum for linear models?**

A: Its convex nature ensures the existence of a single global minimum.

**Q: What is the definition and unit relationship of Root Mean Squared Error (RMSE)?**

A: RMSE is the square root of MSE and shares the same units as the target variable.

**Q: Statistically, minimizing MSE is equivalent to obtaining which estimation parameters?**

A: Minimizing MSE equals finding maximum likelihood estimates under Gaussian noise.

**Q: Write out the normal distribution probability density function for noise $\varepsilon_i$.**

A: $p(y_i|\blacksquare_i) = 1/\sqrt{(2\pi\sigma^2)} * \exp(-(y_i - \blacksquare_i)^2 / 2\sigma^2)$.

**Q: Based on the log-likelihood function derived, what specific term is minimized?**

A: The sum of squared errors term corresponds to the MSE loss being minimized.

**Q: Under the Gaussian noise assumption, why does it make sense to use MSE?**

A: MSE is justified when data errors follow a normal (Gaussian) distribution.

**Q: What is the alternative name for the Mean Absolute Error (MAE)?**

A: Mean Absolute Error (MAE) is also known as L1 Loss or Mean Absolute Deviation.

**Q: What is the formula for L_MAE?**

A: L_MAE = (1/n) Σ |y_i - ■_i|.

**Q: How does the robustness of MAE compare to MSE?**

A: MAE is more robust to outliers than MSE because it applies linear penalties.

**Q: What specific characteristic of |x| is illustrated in the graph?**

A: The |x| graph forms a V-shape with a sharp vertex at zero.

**Q: What is the practical consequence of the derivative discontinuity for MAE?**

A: The discontinuous derivative at zero causes unstable gradients in optimization.

**Q: Explain the core difference between the L1 Loss and L2 Loss.**

A: L1 uses absolute errors while L2 squares them, making L2 more sensitive to large deviations.

**Q: Which loss function would generally be preferred if a dataset contains many outliers?**

A: MAE is preferred when datasets contain extreme outliers due to its linear error handling.

**Q: Why is easy differentiability important for a loss function?**

A: Differentiability allows gradient-based algorithms to optimize parameters effectively.

**Q: If prediction errors are Gaussian, which loss function is justified?**

A: MSE suits models assuming Gaussian-distributed errors.

**Q: What is the relationship between maximizing log-likelihood and minimizing loss?**

A: Maximizing log-likelihood equals minimizing its negative, the loss function.

**Q: What mathematical property of the logarithm simplifies optimization?**

A: The log converts products into sums, simplifying optimization.

**Q: What is the significance of the (n/2)log(2πσ²) term?**

A: The constant term (n/2)log(2πσ²) does not affect minimization.

**Q: What does ε_i represent in y_i = ■_i + ε_i?**

A: ε_i represents random noise or deviation in observed data.

**Q: What is the mathematical definition of the likelihood function L?**

A: L = Π p(y_i|■_i) defines the likelihood under i.i.d. assumption.

**Q: When is it guaranteed that a local minimum is also global?**

A: For convex losses like MSE, any local minimum is also the global minimum.